

Supplementary Appendix
Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab
versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV
Melanoma

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Contents

1 Model Structure and Assumptions

1.1 Health-State Definitions

Table 1: Health-State Definitions

State	Definition
Recurrence-free (RF)	No evidence of locoregional or distant recurrence
Locoregional recurrence (LR)	Confirmed locoregional recurrence without distant metastasis
Distant metastasis (DM)	Confirmed distant metastasis
Death	All-cause mortality

State occupancy is determined by the partitioned survival model (PSM) approach:

$$\begin{aligned}\text{RF}(t) &= \text{RFS}(t) \\ \text{LR}(t) &= \text{DMFS}(t) - \text{RFS}(t) \\ \text{DM}(t) &= \text{OS}(t) - \text{DMFS}(t) \\ \text{Dead}(t) &= 1 - \text{OS}(t)\end{aligned}$$

These are not transition probabilities; they are occupancy probabilities derived from the three survival curves. The constraint $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ is enforced at all times to ensure non-negative state occupancy.

1.2 Model Schematic

[Figure S1: Model schematic — RF → LR → DM → death PSM diagram]

Figure 1: Partitioned survival model structure.

2 Survival Data and Parametric Extrapolation

2.1 Extracted Survival Probabilities

Table 2: Extracted KEYNOTE-942 Survival Probabilities

Month OS (%)	RFS (%)		DMFS (%)		
	Combo	Pembro	Combo	Pembro	Combo
Pembro					
18	80.4	62.2	92.0	73.0	96.0
93.4					
24	78.3	60.0	90.8	70.6	96.0
93.4					
36	73.8	55.6	85.6	65.4	93.8
85.6					
48	72.4	49.1	83.9	65.4	92.2
85.6					
60	68.8	49.1	–	–	92.2
71.3					

Source: Digitized from Kaplan–Meier curves published in Khattak et al. (2026).

2.2 Parametric Model Fit

Table 3: Parametric Model Fit Comparison

Distribution	Parameters	AIC	RMSE	Accepted
<i>OS – Combo arm</i>				
Exponential	1	–	–	–
Weibull	2	–	–	–
Log-normal	2	-50.08	0.0045	Base case
Log-logistic	2	–	–	–
Generalized gamma	3	–	–	–
<i>OS – Pembro arm</i>				
[Full results from survival_model_selection.py]				
<i>r_{DM} – Combo arm</i>				
[Full results]				
<i>r_{DM} – Pembro arm</i>				
[Full results]				
<i>r_{LR} – Combo arm</i>				
[Full results]				
<i>r_{LR} – Pembro arm</i>				
[Full results]				

All 4,096 candidate distribution combinations (4 distributions for 6 curves, excluding over-fitted generalized gamma) satisfied the $RFS \leq DMFS \leq OS$ ordering constraint at all 480 monthly

cycles.

2.3 Observed vs. Fitted Survival Curves

[Figure S2: Observed vs. fitted RFS – Combo and Pembro arms]

Figure 2: Observed versus fitted recurrence-free survival.

[Figure S3: Observed vs. fitted DMFS – Combo and Pembro arms]

Figure 3: Observed versus fitted distant metastasis-free survival.

[Figure S4: Observed vs. fitted OS – Combo and Pembro arms]

Figure 4: Observed versus fitted overall survival.

3 Long-Term Validation

3.1 Long-Term Survival Diagnostics

Table 4: Long-Term Survival Predictions vs. General Population

Year	Combo OS	Pembro OS	GP Survival
5	0.916	0.741	0.945
10	0.825	0.516	0.891
15	0.693	0.392	0.808
20	0.561	0.284	0.704
30	0.214	0.109	0.463
40	0.000	0.000	0.000

[Figure S5: OS vs. general population survival]

Figure 5: Model-predicted OS with general population mortality hazard floor.

3.2 Structural Consistency Checks

Table 5: Structural Consistency Over 480 Monthly Cycles

Check	Result
$\text{RFS} \leq \text{DMFS}$ violations	0
$\text{DMFS} \leq \text{OS}$ violations	0
Negative state probabilities	0
Maximum state-sum deviation	$< 10^{-12}$
Clamp events (GP floor)	0

4 Model Inputs

4.1 Complete Parameter Table

Table 6: Complete Model Input Parameters

Parameter	Base	DSA Range	Source
<i>Survival parameters</i>			
OS μ (combo)	–	–	Fitted
OS σ (combo)	–	–	Fitted
OS μ (pembro)	–	–	Fitted
OS σ (pembro)	–	–	Fitted
r_{DM} μ (combo)	–	–	Fitted
r_{DM} σ (combo)	–	–	Fitted
r_{DM} μ (pembro)	–	–	Fitted
r_{DM} σ (pembro)	–	–	Fitted
r_{LR} μ (combo)	–	–	Fitted
r_{LR} σ (combo)	–	–	Fitted
r_{LR} μ (pembro)	–	–	Fitted
r_{LR} σ (pembro)	–	–	Fitted
<i>Costs (2026 USD)</i>			
Intismeran (per course)	\$200,000	–	Analyst estimate
Pembrolizumab (annual)	\$220,896	–	GoodRx
Sequencing (per course)	\$1,000	–	CMS
Administration (per cycle)	\$500	–	CMS
LR management (monthly)	\$3,000	–	Bensimon 2019
DM management (monthly)	\$12,000	–	Bensimon 2019
AE (incremental, one-time)	\$15,000	–	Bensimon 2019
<i>Utilities</i>			
RF state	0.83	0.03	Beta ($\alpha=129.3$, $\beta=26.5$)
Bensimon 2019, Johnston 2021			
LR state	0.64	0.03	Beta ($\alpha=163.2$, $\beta=91.8$)
Bensimon 2019, Johnston 2021			
DM state	0.55	0.03	Beta ($\alpha=150.7$, $\beta=123.3$)
Bensimon 2019, Zhang 2023			
AE disutility	0.05	–	Fixed (one-time)
Bensimon 2019, Zhang 2023			
<i>Model settings</i>			
Discount rate	3%	–	Standard
Time horizon	40 years	–	Lifetime
Cycle length	1 month	–	Standard
GP mortality	US 2019 Life Table	–	NVSS

Note: DSA ranges: RF 0.80–0.86, LR 0.61–0.67, DM 0.52–0.58 ($\pm 3\%$ of mean). Cost sources: Bensimon 2019, Zhang 2023.

5 Additional Results

5.1 Cost Decomposition

Table 7: Cost Decomposition by Component

Cost component	Combo	Pembro	Incremental
Intismeran	—	—	—
Pembrolizumab	—	—	—
Sequencing	—	—	—
AE/Administration	—	—	—
LR management	—	—	—
DM management	—	—	—
Total	—	—	—

5.2 QALY Decomposition

Table 8: QALY Decomposition by Health State

Health state	Combo	Pembro	Incremental
RF	—	—	—
LR	—	—	—
DM	—	—	—
Total	—	—	—

6 Structural Scenario Analyses

Table 9: Full Scenario Analysis Results

Scenario	Δ Cost	Δ QALY	ICER	NMB @ \$150K	Source
Base case (GP-constrained)	—	—	—	—	—
Unconstrained (log-normal)	—	—	—	—	Methodological
Weibull OS	—	—	—	—	—
GP floor 48 months	—	—	—	—	—
GP floor 72 months	—	—	—	—	—
10-year horizon	—	—	—	—	—
20-year horizon	—	—	—	—	—
Waning v2 (5–10y)	—	—	—	—	—
Waning v2 (5–15y)	—	—	—	—	—
No direct OS benefit	—	—	—	—	—
Pricing: \$100K/course	—	—	—	—	—
Pricing: \$500K/course	—	—	—	—	—

7 Probabilistic Sensitivity Analysis

7.1 PSA Distributions and Covariance

Table 10: PSA Parameter Distributions

Parameter	Distribution	Source
Survival (μ, σ)	Bivariate normal (joint)	NLS covariance
Costs	Gamma (CV=0.2)	Standard
Utilities	Beta (SE=0.03)	Standard
Discount rate	Fixed	Standard

7.2 PSA Convergence

[Figure S6: PSA convergence – mean ICER at 100/500/1000/5000 iterations]

Figure 6: PSA convergence assessment.

7.3 Cost-Effectiveness Plane

[Figure S7: Cost-effectiveness plane (1,000 PSA iterations)]

Figure 7: Probabilistic sensitivity analysis: incremental cost versus incremental QALY.

8 CHEERS 2022 Checklist

Table 11: CHEERS 2022 Health Economic Evaluation Reporting Checklist

#	Item	Reported	Section
1	Title	Yes	p.1
2	Abstract	Yes	p.2
3	Background/Objectives	Yes	Introduction
4	Health economic analysis plan	Yes	Methods
5	Study population	Yes	Methods
6	Setting and location	Yes	Methods
7	Comparators	Yes	Methods
8	Perspective	Yes	Methods
9	Time horizon	Yes	Methods
10	Discount rate	Yes	Methods
11	Selection of outcomes	Yes	Methods
12	Measurement of outcomes	Yes	Methods
13	Valuation of outcomes	Yes	Methods
14	Measurement and valuation of resources and costs	Yes	Methods
15	Currency, price date, and conversion	Yes	Methods
16	Rationale and description of model	Yes	Methods
17	Analytics and assumptions	Yes	Methods
18	Characterizing heterogeneity	N/A	Single population
19	Characterizing distributional effects	N/A	Single population
20	Characterizing uncertainty	Yes	Methods/Results
21	Approach to engagement with patients and others	N/A	Early CEA
22	Study parameters	Yes	Table 2 / Suppl. D
23	Summary of main results	Yes	Results
24	Effect of uncertainty	Yes	PSA/DSA
25	Effect of engagement with patients and others	N/A	Early CEA
26	Study findings, limitations, generalizability	Yes	Discussion
27	Source of funding	Yes	Title page
28	Conflicts of interest	Yes	Title page